

a Association between diagram complexity and forecasting accuracy

Complexity metric (C)	β_1 (C)	β_2 (GenAI)	β_3 ($C \times \text{GenAI}$)	Human slope	GenAI slope
Number of paths	0.002 [−0.027, 0.030]	0.090*** [0.047, 0.134]	−0.020 [−0.056, 0.017]	0.002 [−0.027, 0.030]	−0.018 [−0.041, 0.005]
Maximum path length	0.006 [−0.027, 0.038]	0.082*** [0.035, 0.128]	−0.007 [−0.052, 0.038]	0.006 [−0.027, 0.038]	−0.001 [−0.032, 0.031]
Number of latent variables	0.015 [−0.024, 0.055]	0.087*** [0.038, 0.135]	−0.026 [−0.078, 0.026]	0.015 [−0.024, 0.055]	−0.011 [−0.045, 0.024]

b GenAI–human accuracy differences adjusting for complexity

Complexity metric (C)	γ (GenAI, no C)	γ' (GenAI, with C)	$\gamma' - \gamma$
Number of paths	0.082*** [0.037, 0.128]	0.085*** [0.041, 0.130]	0.003 [−0.014, 0.020]
Maximum path length	0.082*** [0.037, 0.128]	0.079** [0.031, 0.128]	−0.003 [−0.022, 0.016]
Number of latent variables	0.082*** [0.037, 0.128]	0.079** [0.027, 0.131]	−0.003 [−0.036, 0.029]